

```
In [1]: import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\tqdm\aut
o.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets.
See https://ipywidgets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm
```

```
In [2]: url = r"D:\fsds\projects\datasets\eda automization using ollama\titanic_ dataset
df = pd.read_csv(url)
```

```
In [3]: df
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599 7
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803 5
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536 1
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053 3
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607 2
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369 3
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

891 rows × 12 columns

In [4]: `print(df.describe())`

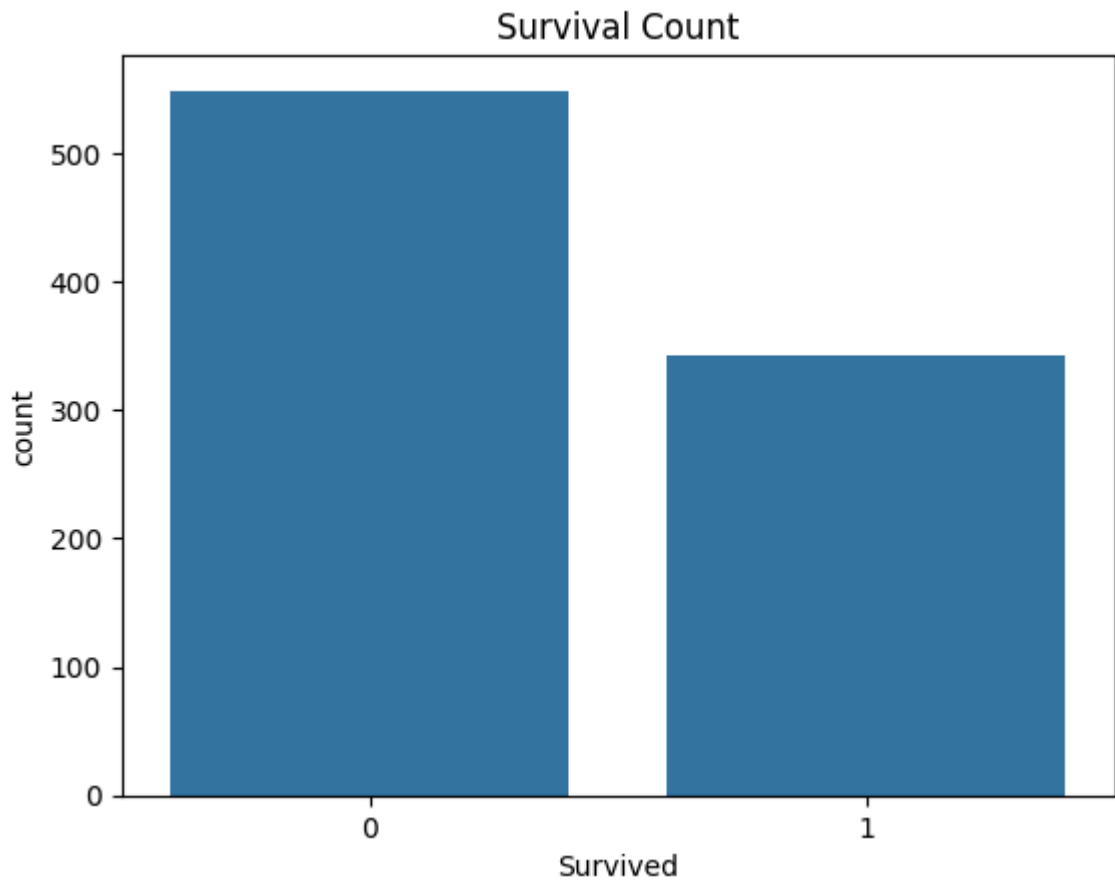
	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
In [5]: print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
  PassengerId      0
  Survived        0
  Pclass          0
  Name            0
  Sex             0
  Age            177
  SibSp           0
  Parch           0
  Ticket          0
  Fare            0
  Cabin          687
  Embarked        2
dtype: int64
```

```
In [6]: sns.countplot(x="Survived", data=df)
plt.title("Survival Count")
plt.show()
```



```
In [7]: import ollama

def generate_insights(df_summary):
    prompt = (
        f"Analyse the following dataset summary and provide insights:\n{df_summary}"
    )
    response = ollama.chat(
        model="gemma", messages=[{"role": "user", "content": prompt}]
    )
    return response["message"]["content"]

summary = df.describe().to_string()
insights = generate_insights(summary)
print("\n * AI Generated Insights:\n", insights)
```

```

* AI Generated Insights:
## Insights from Dataset Summary:

**Passenger Demographics:**

* The dataset includes data on 891 passengers.
* Most passengers (89%) survived (Survived = 0.3838).
* The average age of passengers is 29.7 years.
* Most passengers traveled in 3rd class (64%).

**Passenger Characteristics:**

* About half of the passengers had no siblings or parents traveling with them (SibSp and Parch = 0).
* The average fare paid was 32.20.

**Survival Probabilities:**

* The average survival rate for 1st class passengers is 100%, while it is 28% for 3rd class passengers.
* Age appears to have a slight influence on survival, with younger passengers having a slightly higher survival rate (mean age of survivors = 28.0 vs 44.6).

**Fare and Class Influence:**

* Passengers in 1st class paid significantly more than those in 3rd class (median fare: 14.45 vs 7.91).
* There is a correlation between fare and survival, with higher fares associated with a higher probability of survival.

**Additional Observations:**

* There are outliers in the data, with some passengers paying extremely high fares (maximum fare = 512.32).
* The distribution of Age and Fare appears to be slightly skewed towards the lower end.

**Potential Analysis:**

* Analyze the relationship between passenger class and survival rate.
* Investigate the influence of age on survival probability.
* Explore the correlation between fare and survival, controlling for class.
* Identify potential risk factors for drowning based on passenger characteristics.

```

```

In [8]: import gradio as gr

def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)

    return insights
demo = gr.Interface(fn=eda_analysis, inputs='file', outputs='text', title='AI Po
demo.launch()

```

```

* Running on local URL: http://127.0.0.1:7860
* To create a public link, set `share=True` in `launch()`.

```

AI Powered EDA with Gemma

file

Drop File Here
- or -
Click to Upload

Clear

Submit

output

Out[8]:

```
Traceback (most recent call last):
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\gradio\queueing.py", line 625, in process_events
    response = await route_utils.call_process_api(
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\gradio\route_utils.py", line 322, in call_process_api
    output = await app.get_blocks().process_api(
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\gradio\blocks.py", line 2220, in process_api
    result = await self.call_function(
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\gradio\blocks.py", line 1731, in call_function
    prediction = await anyio.to_thread.run_sync( # type: ignore
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\to_thread.py", line 56, in run_sync
    return await get_async_backend().run_sync_in_worker_thread(
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\_backends\_asyncio.py", line 2177, in run_sync_in_worker_thread
    return await future
           ^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\anyio\_backends\_asyncio.py", line 859, in run
    result = context.run(func, *args)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\Users\bhara\AppData\Local\Programs\Python\Python312\Lib\site-packages\gradio\utils.py", line 904, in wrapper
    response = f(*args, **kwargs)
              ^^^^^^^^^^^^^^^^^^
  File "C:\Users\bhara\AppData\Local\Temp\ipykernel_43184\3766487616.py", line 4, in eda_analysis
    df = pd.read_csv(file.name)
         ^^^^^^^^^^^
AttributeError: 'NoneType' object has no attribute 'name'
```

```
In [9]: import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import ollama

# Function to Perform EDA and Generate Visualizations
def eda_analysis(file_path):
    df = pd.read_csv(file_path)

    # Fill missing values with median for numeric columns
    for col in df.select_dtypes(include=["number"]).columns:
        df[col].fillna(df[col].median(), inplace=True)

    # Fill missing values with mode for categorical columns
    for col in df.select_dtypes(include=["object"]).columns:
        df[col].fillna(df[col].mode()[0], inplace=True)

    # Data Summary
    summary = df.describe(include="all").to_string()
```

```

# Missing Values
missing_values = df.isnull().sum().to_string()

# Generate AI Insights
insights = generate_ai_insights(summary)

# Generate Data Visualizations
plot_paths = generate_visualizations(df)

return (
    f"\n Data Loaded Successfully!\n\n Summary:\n{summary}\n\n Missing Value
    plot_paths,
)

# AI-Powered Insights using Mistral-7B (Ollama)
def generate_ai_insights(df_summary):
    prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}"
    response = ollama.chat(
        model="mistral", messages=[{"role": "user", "content": prompt}]
    )
    return response["message"]["content"]

# Function to Generate Data Visualizations
def generate_visualizations(df):
    plot_paths = []

    # Histograms for Numeric Columns
    for col in df.select_dtypes(include=["number"]).columns:
        plt.figure(figsize=(6, 4))
        sns.histplot(df[col], bins=30, kde=True, color="blue")
        plt.title(f"Distribution of {col}")
        path = f"{col}_distribution.png"
        plt.savefig(path)
        plot_paths.append(path)
        plt.close()

    # Correlation Heatmap (only numeric columns)
    numeric_df = df.select_dtypes(include=["number"])
    if not numeric_df.empty:
        plt.figure(figsize=(8, 5))
        sns.heatmap(
            numeric_df.corr(), annot=True, cmap="coolwarm", fmt=".2f", linewidth
        )
        plt.title("Correlation Heatmap")
        path = "correlation_heatmap.png"
        plt.savefig(path)
        plot_paths.append(path)
        plt.close()

    return plot_paths

# Gradio Interface
demo = gr.Interface(
    fn=eda_analysis,
    inputs=gr.File(type="filepath"),
    outputs=[gr.Textbox(label="EDA Report"), gr.Gallery(label="Data Visualizatio

```



```
title="📊 LLM-Powered Exploratory Data Analysis (EDA)",  
description="Upload any dataset CSV file and get automated EDA insights with  
)  
  
# Launch the Gradio App  
demo.launch(share=True)
```

* Running on local URL: <http://127.0.0.1:7861>

Could not create share link. Missing file: C:\Users\bhara\.cache\huggingface\grad
io\frpc\frpc_windows_amd64_v0.3.

Please check your internet connection. This can happen if your antivirus software
blocks the download of this file. You can install manually by following these ste
ps:

1. Download this file: https://cdn-media.huggingface.co/frpc-gradio-0.3/frpc_windows_amd64.exe
2. Rename the downloaded file to: frpc_windows_amd64_v0.3
3. Move the file to this location: C:\Users\bhara\.cache\huggingface\gradio\frpc



LLM-Powered Exploratory Data Analysis (EDA)

Upload any dataset CSV file and get automated EDA insights with AI-powered
analysis and visualizations.

 file_path


Drop File Here
- or -
Click to Upload

Clear

Submit

EDA Report

Out[9]:

C:\Users\bhara\AppData\Local\Temp\ipykernel_43184\3769162220.py:14: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

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df[col].fillna(df[col].median(), inplace=True)
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df[col].fillna(df[col].mode()[0], inplace=True)
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